

MSTAR PRACTICE WORKSHEET**Lesson 4-2: Relate Fractions, Decimals, and Percentages**

Grade 6 Mathematics · Standard 6.AT.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: C**Convert $\frac{3}{5}$ to a percent.

- ☐ A 35%
- ☐ B 53%
- ☒ C 60%
- ☐ D 65%

$5 \times 20 = 100$, so $3 \times 20 = 60$. That makes $\frac{3}{5} = \frac{60}{100} = 0.6 = 60\%$.

Question 2 **SELECTED RESPONSE — CORRECT: C**Convert $\frac{7}{20}$ to a percent.

- ☐ A 14%
- ☐ B 20%
- ☒ C 35%
- ☐ D 70%

$20 \times 5 = 100$, so $7 \times 5 = 35$. That makes $\frac{7}{20} = \frac{35}{100} = 0.35 = 35\%$.

Question 3 SELECTED RESPONSE — CORRECT: B

Which decimal is equivalent to 45%?

- ☐ A 0.045
- ☒ B 0.45
- ☐ C 4.5
- ☐ D 45.0

45% means $45/100$, and $45/100$ written as a decimal is 0.45 (forty-five hundredths).

Question 4 SELECTED RESPONSE — CORRECT: A

What fraction is equal to 0.75?

- ☒ A $3/4$
- ☐ B $3/5$
- ☐ C $7/5$
- ☐ D $7/10$

$0.75 = 75/100$. Simplify by dividing both by 25: $75 \div 25 = 3$ and $100 \div 25 = 4$. So $0.75 = 3/4$.

Question 5 SELECTED RESPONSE — CORRECT: C

A game shows your score as $7/8$ complete. What percent is that?

- ☐ A 75%
- ☐ B 78%
- ☒ C 87.5%
- ☐ D 88%

8 does not scale to 100 by a whole number, so divide: $7 \div 8 = 0.875$, and $0.875 \times 100 = 87.5\%$. That is 87.5 out of 100.

Question 6

SELECTED RESPONSE — CORRECT: B

Your arcade scoreboard reads 0.25 of the level cleared. What percent and fraction match this score?

- ☐ A 2.5% and $\frac{1}{4}$
- ☒ B 25% and $\frac{1}{4}$
- ☐ C 25% and $\frac{1}{5}$
- ☐ D 250% and $\frac{1}{40}$

Move the decimal two places: $0.25 = 25\%$. As a fraction, $0.25 = \frac{25}{100} = \frac{1}{4}$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: B

What is the fraction $\frac{7}{20}$ written as a decimal and as a percent?

- ☐ A 0.035 and 3.5%
- ☒ B 0.35 and 35%
- ☐ C 0.35 and 3.5%
- ☐ D 0.7 and 70%

$\frac{7}{20} = (7 \times 5) / (20 \times 5) = \frac{35}{100} = 0.35 = 35\%$.

- **A:** The decimal point is one place too far left. 7 divided by 20 is 0.35.
- **C:** The decimal is right but the percent is not. 0.35 is 35 hundredths, so it is 35%.
- **D:** That is $\frac{7}{10}$, not $\frac{7}{20}$. Scale to hundredths: $\frac{7}{20}$ equals $\frac{35}{100}$.

Part B — correct answer: A

Which mathematical reasoning justifies the conversion in Part A?

- ☒ A Multiplying both numerator and denominator by 5 yields the equivalent fraction $\frac{35}{100}$, which equals 0.35 and 35%.
- ☐ B Moving the decimal point in 7.20 one place left gives 0.72.
- ☐ C Adding 15 to both terms gives $\frac{22}{35}$, which equals 35%.
- ☐ D Dividing 20 by 7 gives approximately 2.85, so the percent is 35%.

Scaling the denominator to 100 by multiplying by $\frac{5}{5}$ preserves equivalence and directly reveals the rate per hundred.

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Spot the Decimal-Percent Error

Elena wrote that 0.08 is equal to 80% because 'there is an 8 in the number.'

Explain Elena's misconception and determine the correct percent value.

Model answer: $0.08 = 8\%$. Elena confused hundredths with tenths; 80% is 0.80 ($80/100$), whereas 0.08 is 8 hundredths ($8/100 = 8\%$).

2 points	Student clearly identifies that 0.08 has 8 in the hundredths place ($8/100 = 8\%$) and explains that 80% is 0.80 ($80/100$).
1 point	Student provides the correct answer (8%) but provides an incomplete or vague explanation of place value.
0 points	Student agrees with Elena or gives an incorrect response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

A student says $3/8$ is greater than 40% because 3 and 8 are bigger numbers than 4 and 0. Is the student correct? Convert both values to the same form and explain.

Model answer: Bigger-looking digits do not mean a bigger value — the two amounts have to be in the same form before you compare them. Converting the fraction gives $3 \div 8 = 0.375$, which is 37.5%. Since 37.5% is less than 40%, the student is not correct: $3/8$ is the smaller value.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.